



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI AYYAPPA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

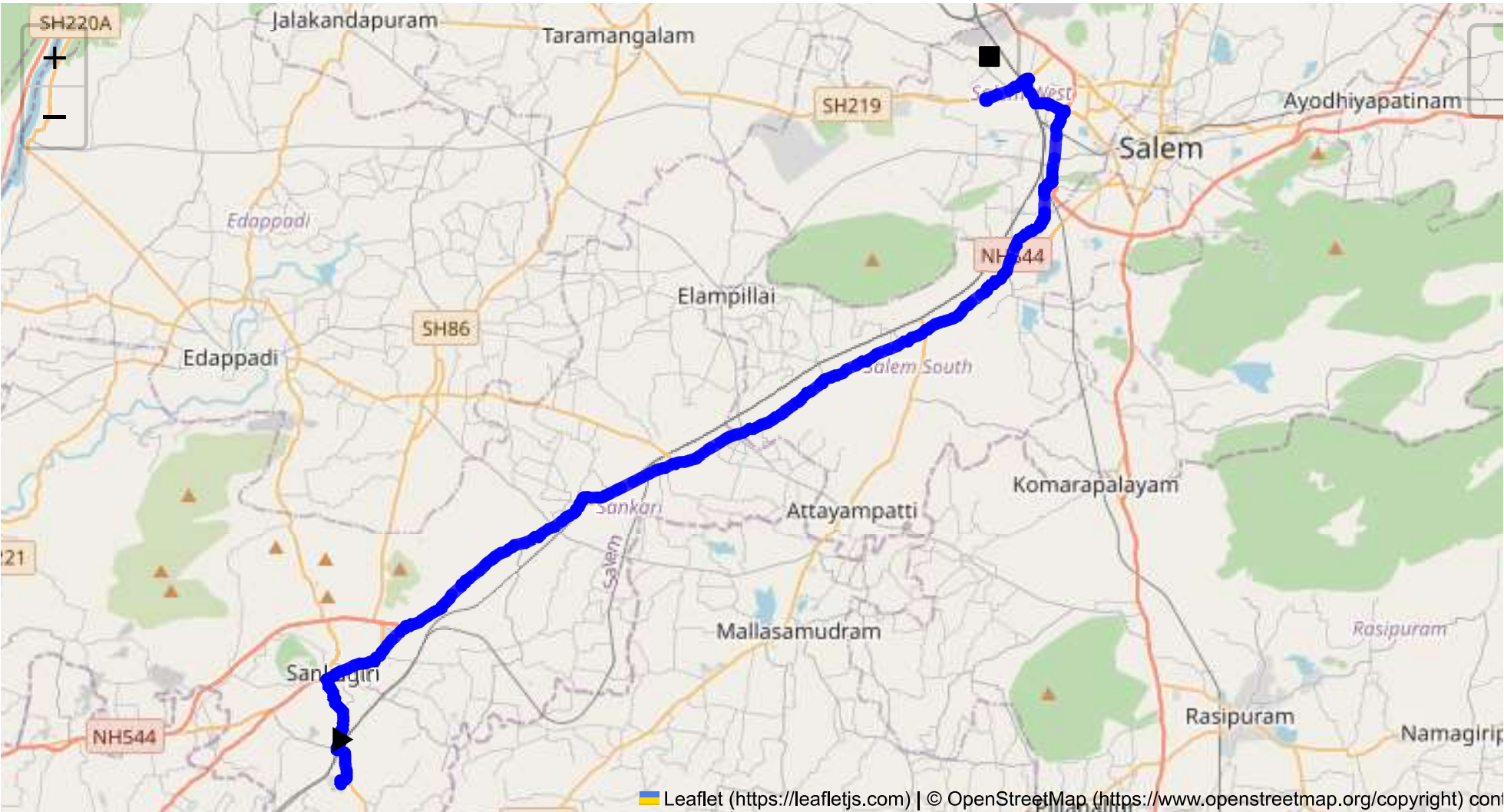
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 46.08 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.671329, 78.097878)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 123/80, Palanisamy Nagar in Salem is approximately 46.08 kilometers. It travels mainly on local roads and highways, likely including SH 86 and NH 544. These roads are typically two-lane and cater to both local traffic and long-distance travel across Tamil Nadu.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences tropical weather, with heavy rainfall from the northeast monsoon between October and December. During this period, roads may become slippery and prone to flooding, particularly in low-lying areas. High temperatures during summer can also affect asphalt conditions, potentially leading to softened road surfaces.

3. Analysis of Traffic Patterns

Traffic congestion is common in urban areas, particularly near Salem during peak hours (8-10 AM and 5-7 PM). The roads may experience moderate to heavy traffic due to local commuters and commercial vehicles. Keep an eye on intersections and city exits, where congestion is more likely.

4. Assessment of Road Quality and Infrastructure

The highways like NH 544 are generally well-maintained, whereas some local roads may suffer from potholes and uneven surfaces, particularly after the rainy season. Road signage and lighting may vary, so caution is advised during night travel or in less maintained stretches.

5. Suggestions for Alternative Routes for Emergencies

Alternative routes might include moving through SH 20 or other nearby state highways to bypass heavy congestion areas or road closures. Familiarity with local diversions and updated GPS services could provide real-time alternatives.

6. Summary of Local Regulations Affecting Hazardous Material Transport

The transport of hazardous materials in Tamil Nadu is regulated under national guidelines that require specific permits, adherence to specific routes, and restricted timings, often avoiding peak traffic and school hours to mitigate risk.

7. Overview of Historical Incidents

While specific historical incidents involving heavy vehicles along this route are sparse, Tamil Nadu generally records issues related to vehicle overload and mismanagement. Past incidents often involved overturned vehicles due to sharp turns and unanticipated maneuvers.

8. Environmental Considerations and Sensitive Areas

Potentially sensitive areas along the route include agricultural lands and any water bodies near Salem. Operators must be cautious to prevent contamination, especially near water resources.

9. Analysis of Communication Coverage

Communication infrastructure is generally good along highways but may weaken in rural stretches between towns. Service providers like Airtel, Jio, and BSNL generally offer the best coverage, though sporadic dead zones may still exist.

10. Estimated Emergency Response Times

Emergency response varies but generally expect a 30-45 minute response time in rural segments, improving to 15-30 minutes as you near urban Salem. The key is knowing local hospitals and emergency services in advance.

12. Overall Summary of Risk Assessment

- **Traffic & Congestion:** High risks near urban areas, especially during peak hours.
- **Road Quality & Weather:** Wet and slippery conditions during monsoon; summer can affect road surface.
- **Regulations & Safety:** Ensure compliance with local hazardous materials transport laws.
- **Infrastructure:** Generally good on highways, variable on smaller roads.
- **Emergency Preparedness:** Have an awareness of emergency contacts, hospitals, and alternative routes. Overall, the route is manageable with planned precautions and consistent monitoring of weather updates and real-time traffic information. Regular checks with respect to vehicle maintenance and driver preparedness can further mitigate risks associated with transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45004, 77.87206	15 KM/Hr
5	Turn	Medium	11.45028, 77.87202	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	Medium	11.47410, 77.86845	30 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.61736, 78.10588	30 KM/Hr
11	Turn	Medium	11.61753, 78.10608	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	Medium	11.64007, 78.11794	30 KM/Hr
13	Turn	Medium	11.64020, 78.11786	30 KM/Hr
14	Turn	Medium	11.64497, 78.12028	30 KM/Hr
15	Turn	Medium	11.64503, 78.12034	30 KM/Hr
16	Turn	High	11.66458, 78.12303	15 KM/Hr
17	Turn	High	11.66477, 78.12298	15 KM/Hr
18	Turn	High	11.66719, 78.12442	15 KM/Hr
19	Turn	Medium	11.66762, 78.12417	30 KM/Hr
20	Turn	Medium	11.67015, 78.11449	30 KM/Hr
21	Turn	Medium	11.67016, 78.11440	30 KM/Hr
22	Turn	Medium	11.67023, 78.11431	30 KM/Hr
23	Turn	Medium	11.67664, 78.11117	30 KM/Hr
24	Turn	Medium	11.67716, 78.11079	30 KM/Hr
25	Turn	High	11.67840, 78.11186	15 KM/Hr
26	Turn	Medium	11.67849, 78.11179	30 KM/Hr
27	Turn	High	11.67851, 78.11172	15 KM/Hr
28	Blind Spot	Blind Spot	11.67153, 78.09716	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
22	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
23	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
24	hospital	M.J. Hospital	11.6689574, 78.1209471	30 km/h	Medium
25	police	Suramangalam Police Station	11.6708007, 78.1168863	30 km/h	Medium
26	police	Government Railway Police Station	11.6712204, 78.1130289	30 km/h	Medium
27	hospital	Janani Medical Centre Hospital	11.675896, 78.104826	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
16	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
17	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
19	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
20	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
21	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45004, 77.87206



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47410, 77.86845



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61736, 78.10588



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61753, 78.10608



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64007, 78.11794



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64020, 78.11786



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64497, 78.12028



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64503, 78.12034



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66458, 78.12303



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66477, 78.12298



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66719, 78.12442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66762, 78.12417



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67015, 78.11449



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67016, 78.11440



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67023, 78.11431



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67664, 78.11117



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67716, 78.11079



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67840, 78.11186



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67849, 78.11179



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67851, 78.11172



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.67153, 78.09716

